SOFTWARE TESTING PRACTICAL REPORT

PRACTICAL REVIEW — PR-14

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Enrollment Number	2504030100419
Programme	BCA — Semester 3
Practical	PR-14 — Selenium Web Automation: Login and Cart Validation
Tools	Python, Selenium WebDriver, Chrome, PyCharm, SauceDemo

Prepared from the five screenshots supplied for PR-14.

1. Aim

To automate the SauceDemo shopping workflow using Selenium WebDriver: open the website, enter valid credentials, validate login, add a product to the cart, open the cart, and verify the cart page.

2. Objectives

• Launch Chrome and navigate to the SauceDemo website.

• Locate and fill the username and password fields.

• Click Login and confirm navigation to the inventory page.

• Add the Sauce Labs Backpack to the shopping cart.

• Open the cart and validate the cart URL.

• Close the browser after test execution.

3. Tools and Requirements

Tool	Purpose
Python	Runs the automation script.
Selenium WebDriver	Finds page elements and automates browser interactions.
Chrome	Browser under test.
PyCharm	IDE used to edit and run tutorial14.py.
SauceDemo	Demo shopping website used for login and cart testing.

4. Test Procedure

• Create tutorial14.py in the PyCharm project and import Selenium WebDriver, By, WebDriverWait, and expected_conditions.

• Start Chrome, maximize the window, and create a 10-second explicit wait.

• Open https://www.saucedemo.com/ and enter standard_user and secret_sauce.

• Click the login button and wait until the URL contains “inventory”.

• Click the Add to cart button for Sauce Labs Backpack.

• Click the shopping cart icon and check that the URL contains “cart”.

• Run driver.quit() in the finally block to close the browser.

5. Program Code

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

# Start Chrome
driver = webdriver.Chrome()

# Maximize browser
driver.maximize_window()

# Create wait object
wait = WebDriverWait(driver, 10)

try:
# 1. Browser Navigation
```

```
driver.get("https://www.saucedemo.com/")
print("Website opened")

# 2. Form Handling - Username
username = wait.until(
EC.visibility_of_element_located((By.ID, "user-name"))
)
username.send_keys("standard_user")

# 3. Form Handling - Password
password = driver.find_element(By.ID, "password")
password.send_keys("secret_sauce")

# 4. Click Login
driver.find_element(By.ID, "login-button").click()

# 5. Validation
wait.until(EC.url_contains("inventory"))
assert "inventory" in driver.current_url
print("Login Validation: PASS")

# 6. Web Element Interaction
wait.until(
EC.element_to_be_clickable(
(By.ID, "add-to-cart-sauce-labs-backpack")
)
).click()
print("Product Selection: PASS")

# 7. Navigate to Cart
driver.find_element(By.CLASS_NAME, "shopping_cart_link").click()

# 8. Validate Cart
assert "cart" in driver.current_url
print("Cart Validation: PASS")

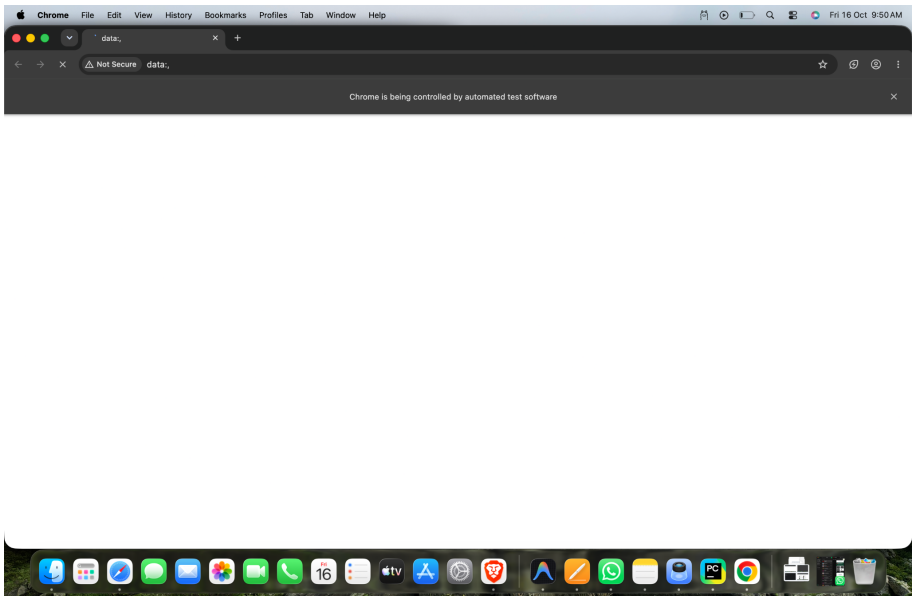
finally:
# Close browser
driver.quit()

print("Tutorial 14 Completed Successfully")
```

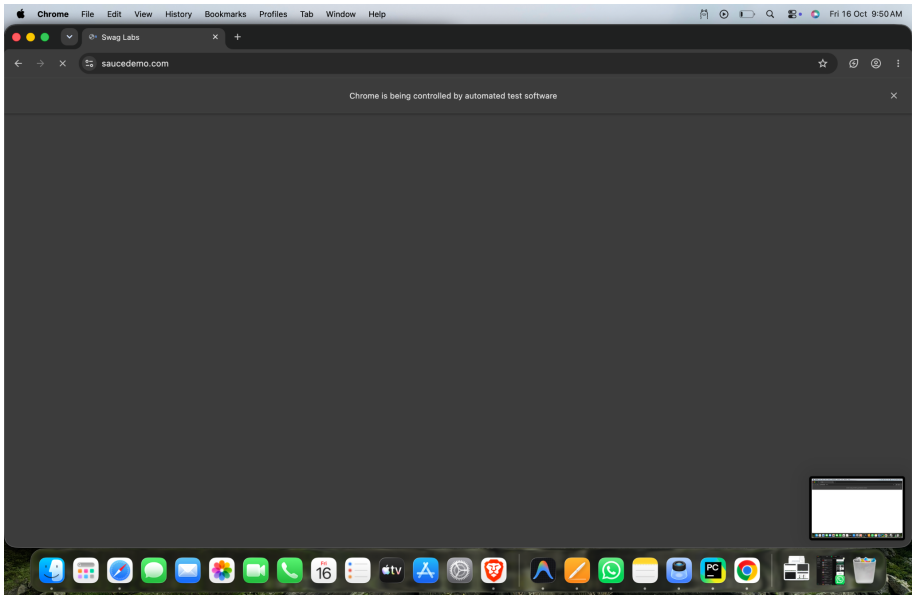
6. Screenshot Evidence

Screenshots document the browser navigation and inventory page, along with the Tutorial 14 source code. The supplied Run panel still displays the st12.py output, so it does not confirm the PR-14 console results.

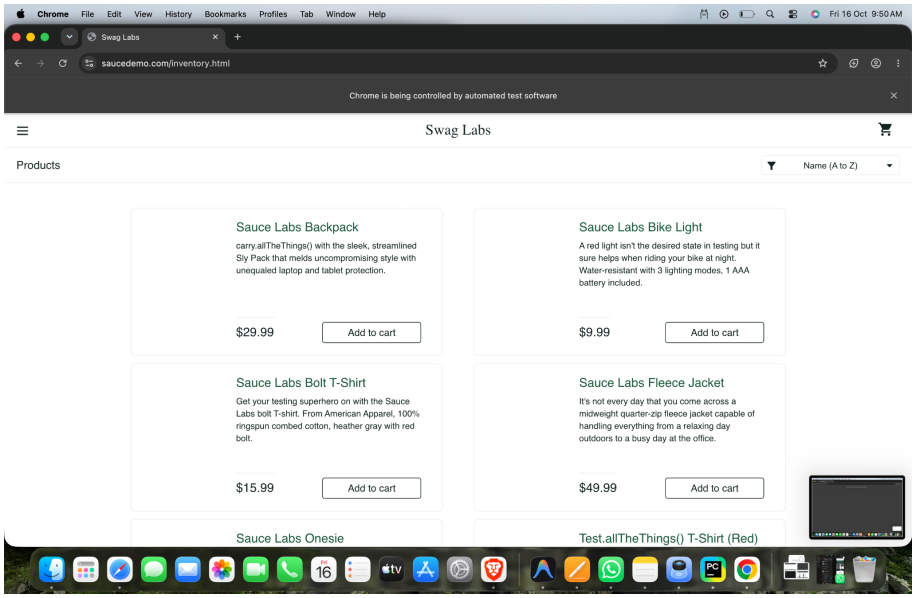
01 Browser launched to an empty data: page (initial navigation state)



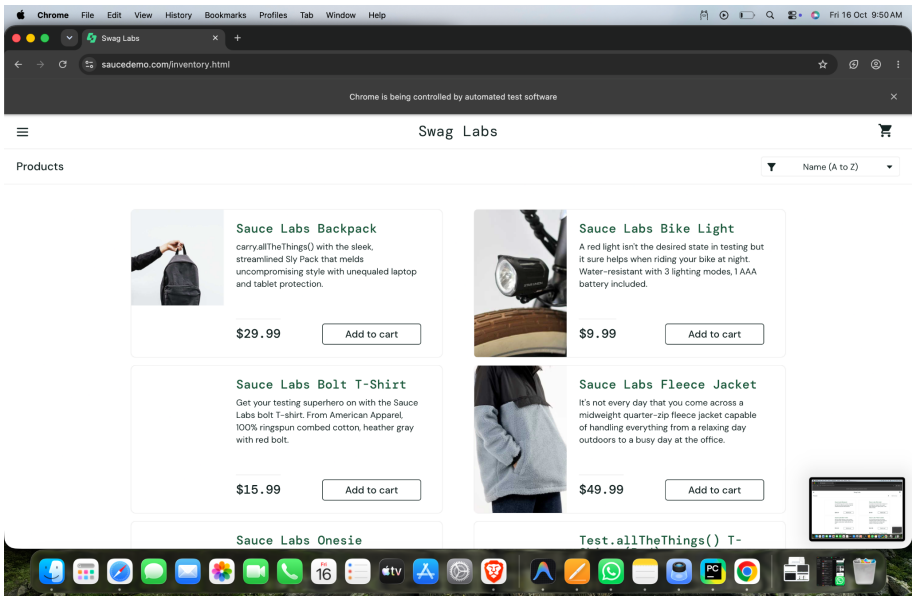
02 SauceDemo site loading in Chrome



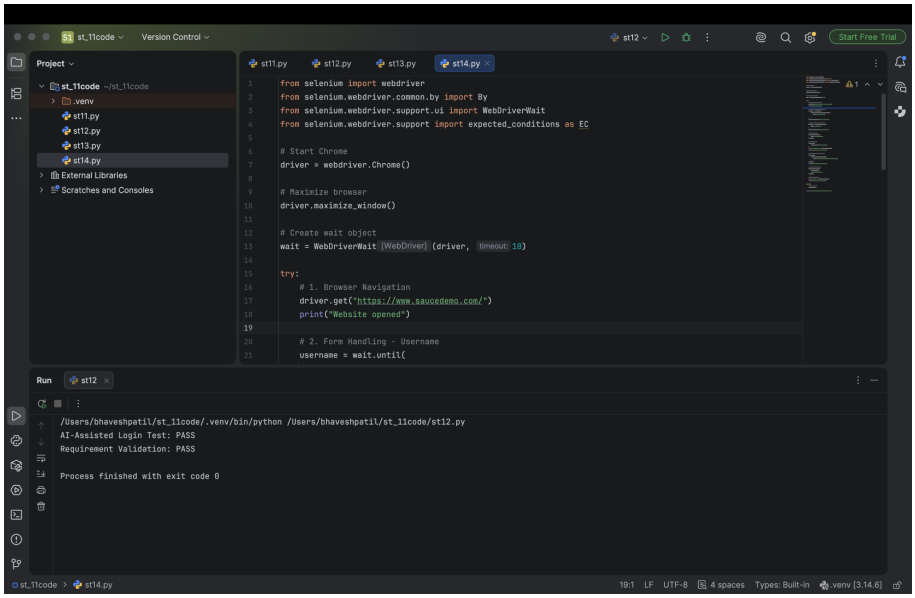
03 Inventory page displayed after successful login



04 Product list visible with product cards



05 PyCharm Tutorial 14 source code and project run panel



7. Test Case Summary

ID	Test	Expected result	Evidence / status
TC-01	Open website	SauceDemo login page loads.	Navigation screenshots supplied; first capture shows data: page.
TC-02	Login	Valid credentials open inventory.	Inventory screenshots supplied.
TC-03	Add product	Backpack is added to cart.	Code includes click action; no cart badge/result screenshot.
TC-04	Open cart	Cart page opens.	Code includes navigation; cart page screenshot not supplied.
TC-05	Validate cart URL	URL contains cart.	Assertion in code; PR-14 console output not supplied.

8. Observations

• The supplied screenshots show Chrome reaching the SauceDemo inventory page.

• The first screenshot captures a blank data: page during an initial navigation state.

• The source-code screenshot shows tutorial14.py with the login, product selection, and cart validation steps.

• The visible Run console is still labelled st12 and reports Tutorial 12 results. A PR-14 run screenshot is required to verify the final PASS messages.

9. Result

The Tutorial 14 automation script is structured to test login, product selection, and cart navigation on SauceDemo. The screenshots show the inventory page and source code, but the final cart assertions cannot be confirmed from the supplied evidence alone.

10. Conclusion

Selenium WebDriver can automate common e-commerce actions, validate page navigation, and use assertions to check expected behavior.

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